

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda

PROFESSIONAL SUMMARY

Backend engineer with 4+ years building production SaaS products at scale. Led zero-downtime migration of 20+ applications to Firebase/GCP, driving adoption across engineering and operations teams. Experience integrating AI services (OpenAI, Google Vision) into customer-facing products. Prototype and ship rapidly while maintaining quality. Strong technical communicator who thrives at the intersection of engineering depth and strategic customer impact. Proficient in JavaScript/Node.js and REST API system integration.

CORE COMPETENCIES

Node.js & JavaScript

REST API Integration

Firebase & GCP

SaaS Product Development

Cross-Functional Collaboration

Technical Communication

AI/ML Integration

Rapid Prototyping

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Drove adoption of **Firebase/GCP** across 20+ applications, partnering with CTO to design and deliver zero-downtime migration impacting entire production stack
- Built real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub for **REST API integration** across systems
- Developed 3D visualization feature using video overlay and ffmpeg, **increasing buyer conversion**
- Trained operations team** on new Firebase Dashboard and backup procedures, creating documentation for non-technical stakeholders
- Saved company \$5,000/month by migrating all services off AWS — demonstrating ability to **prototype and ship** cost-effective solutions quickly

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Hired via Upwork to execute complex backend migration from Parse/MongoDB to Firebase Firestore and GCP
- Built migration script handling complex processing logic for **system integration** between legacy and modern stacks
- Wrote documentation for engineers on new Firebase SDKs (Node.js and Python) — **cross-functional collaboration** with R&D team

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led optimization of PWA delivering real-time stock prices from Firebase backend — **SaaS product** serving end users
- Built ad-hoc jobs to query 2M+ Firestore records, keeping customer prices current via **REST API integration** with Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js — **rapid prototyping** under tight deadlines

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Full-stack development for US agency with demanding clients — **prototype and ship** under tight timelines
- Built backends for DeFi applications using Web3 and Node.js — demonstrating ability to **grasp novel architectures** quickly
- Delivered React UIs from Figma designs, **cross-functional collaboration** with design and product teams

CodeBits

Jan 2020 – Jul 2021

Software Engineer · Uganda Remote

- Led team of 4 developers building legal tech systems for NGOs — **strategic customer** engagement with FIDA Uganda and LASPNET
- Architected microservices backend using Apache Kafka, Docker, and Kubernetes for case management **SaaS product**
- Built USSD service for legal aid providers via gRPC — **system integration** with mobile carrier APIs
- Developed cross-platform mobile app (React Native/Expo) with Firebase push notifications — **rapid prototyping** to production

PROJECTS

Healthcare WhatsApp Chatbot AI Integration

Integrated **OpenAI** and **Google Vision AI** for prescription OCR via WhatsApp Business API. Demonstrates ability to grasp novel AI/ML problems and ship production solutions.

Node.js, OpenAI API, Google Cloud Vision, WhatsApp Business API

Marketplace Recommendation Engine AI/ML

Built collaborative and content-based filtering engine in MongoDB for service provider matching. Applied **ML techniques** to production recommendation system.

Node.js, MongoDB, Python

Real-time Pub/Sub Sync Pipeline System Integration

Built live MongoDB ↔ Firestore sync pipeline supporting parallel production traffic during migration. **REST API integration** with message queue architecture.

Node.js, Google Pub/Sub, MongoDB, Firestore

EDUCATION

Bachelor of Science in Computer Engineering

2018 – 2022

[Makerere University](#), Kampala, Uganda

SKILLS

Languages: JavaScript, Python, Go, Rust **Backend:** Node.js, Express.js, Firebase Cloud Functions, gRPC

Cloud: GCP, Firebase, AWS (S3, EC2, EBS) **Databases:** Firestore, MongoDB, SQL

Infrastructure: Docker, Kubernetes, Apache Kafka, Pub/Sub **Frontend:** React, React Native, Tailwind, HTML/CSS

AI/ML: OpenAI API, Google Cloud Vision